Web Design Final Exam Project

Project Overview:

For your final web design exam, you are required to create a complete web-based project based on a real-life situation (a case study) that **solves a specific problem** using web technologies.

Your project must include both **front-end (what users see and interact with)** and **back-end (where data is stored and processed)** development skills.

Task to Do:

1. Choose a Real-World Problem

- Think of a real problem in your community, school, or daily life.
- Example: booking a medical appointment, managing school activities, or selling products online…etc
- Create a website that offers a solution to this problem.

2. Design and Build the Website

Use the following technologies:

- **HTML**: To create the structure of your website.
- **CSS**: To make your website look professional and responsive.
- **JavaScript**: To add interactive features and check user inputs.
- **PHP and MySQL**: To connect your forms to a database and store the submitted data.

3. Make Your Website Interactive

- Add features like show/hide menus, live feedback, or image sliders etc
- Use **JavaScript** to check if users enter the correct information in forms (e.g., email format, required fields).

4. Connect to a Database (Backend)

- Create a form (e.g., registration, contact form).
- Use **PHP** to send the data from the form to a **MySQL database**.
- Make sure the database stores the data correctly and securely.

5. Validate and Protect Your Data

- Make sure all required fields are filled.
- Limit incorrect or harmful data from being submitted.
- Use proper data types (text, number, date, etc.).

Part B: Documentation

- You are required to **prepare and print your full project documentation** as a **hardcopy**.
- The printed report must be **well-organized**, **typed (not handwritten)**, and **properly formatted**.
- Include the following in your printed documentation:
    1. Cover page with your name, registration number, course, and project title
    2. Table of contents
    3. All sections outlined in the "Project Documentation" part of this brief
    4. Screenshots of your system (interface, forms, database, etc.)

What to Include in Your Report (Documentation)?

Your final report should include the following sections:

1. Project Introduction

- **Title**: (e.g., Online Bookstore, Event Management System)
- **Case Study**: Who is this system built for?
- **Purpose**: What problem does it solve?
- **Technologies Used**: (HTML5, CSS3, JavaScript, PHP, MySQL, etc.)

2. Problem Statement

- Explain the problem you're solving.
- Who are the users?
- What features will help them?

3. System Requirements

- **Software**: Browser, Code Editor (VS Code), Server (XAMPP/WAMP)
- (Hardware details are optional)

4. System Design

- Show how the user will use your system (a diagram or flowchart).
- List your pages (Home, About, Contact, etc.).
- Describe how users move through your site (menus, links).

5. Implementation (Coding Part)

Describe what you did in:

- **HTML/CSS/JS**: Layout, styling, and interactivity
- **PHP**: Handling form submissions and database actions
- **Database**: Tables created, fields, and how data is stored

6. Database Design

- Draw your **ER Diagram** (if applicable)
- List all **tables**, their **columns**, and data types
- Explain relationships between tables (e.g., user → orders)

7. Testing

- How did you test your project?
  - Manually tested forms, login, errors
  - Checked validation
  - Logged in as different users (if applicable)
- Mention any **bugs** and how you fixed them

8. Challenges Faced

- What problems did you face during development?
- How did you solve them?

9. Conclusion

- What did you learn?
- What would you improve in the future?

10. Screenshots

- Include clear screenshots of your:
  - Web pages
  - Forms
  - Database
  - User actions (e.g., registration, login)